Describing Aircraft Dynamics

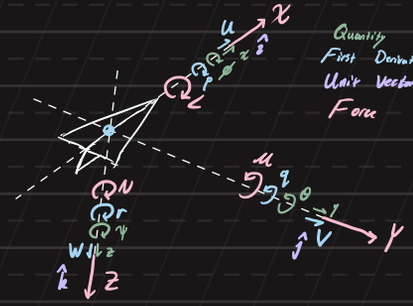
For a rigid body:

$$\vec{F} = \frac{dm\vec{v}}{dt} \approx m\frac{d\vec{v}}{dt}$$ } only for an inertial observer (we can't be accelerating)

$$\vec{M} = \frac{d}{dt}\underline{I}(\omega)$$ } $\underline{I}$ is a Tensor? ← What is a tensor

We can choose our axis s.t. the diagonal terms are 0 (by Symmetry). We choose the aircraft at equilibrium as the axes.

Coordinate Axes



Quantity
First Derivative
Unit Vector
Force

Equations of Motion

$$I = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{xz} & 0 & I_{zz} \end{bmatrix}$$ } We assume vertical Symmetry (i.e. plane including $\hat{i}$ & $\hat{k}$)

$\underline{v} = \langle u, v, w \rangle$
$\underline{\omega} = \langle p, q, r \rangle$
$\underline{F} = \langle X, Y, Z \rangle$
$\underline{M} = \langle L, M, N \rangle$

$$\frac{d\hat{i}}{dt} = \underline{\omega} \times \hat{i} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ p & q & r \\ 1 & 0 & 0 \end{vmatrix} = r\hat{j} - q\hat{k}$$

$$\frac{d\hat{j}}{dt} = \underline{\omega} \times \hat{j} = -r\hat{i} + p\hat{k}$$

$$\frac{d\hat{k}}{dt} = \underline{\omega} \times \hat{k} = q\hat{i} - p\hat{j}$$

$$\frac{d}{dt}[m\underline{v}] = m\left(\frac{du}{dt}\hat{i} + \frac{dv}{dt}\hat{j} + \frac{dw}{dt}\hat{k} + u\frac{d\hat{i}}{dt} + v\frac{d\hat{j}}{dt} + w\frac{d\hat{k}}{dt}\right)$$

$$\frac{d}{dt}[\underline{I}(\omega)] = \left(I_{xx}\frac{dp}{dt} - I_{xz}\frac{dr}{dt}\right)\hat{i} + I_{yy}\frac{dq}{dt}\hat{j} + \left(-I_{xz}\frac{dp}{dt} + I_{zz}\frac{dr}{dt}\right)\hat{k}$$
$$+ (I_{xx}p - I_{xz}r)\frac{d\hat{i}}{dt} + (I_{yy}q)\frac{d\hat{j}}{dt} + (-I_{xz}p + I_{zz}r)\frac{d\hat{k}}{dt}$$

$$\frac{d}{dt}(m\underline{v}) = \underline{F}, \quad \frac{d}{dt}[\underline{I}(\omega)] = \underline{M}$$

Component Form of the governing Differential Equations

Newton's Equations

$$m(\dot{u} - vr + wq) = X$$
$$m(\dot{v} + ur - wp) = Y$$
$$m(\dot{w} - uq + vp) = Z$$

Euler's Equations

$$[I_{xx}\dot{p} - I_{xz}\dot{r} + (I_{zz} - I_{yy})qr - I_{xz}pq] = L$$
$$[I_{yy}\dot{q} + (I_{xx} - I_{zz})pr + I_{xz}(p^2 - r^2)] = M$$
$$[I_{zz}\dot{r} - I_{xz}\dot{p} + (I_{yy} - I_{xx})pq + I_{xz}qr] = N$$

$$d\underline{H}_G = (\underline{r} \times \underline{v})dm$$
$$\underline{v} = \underline{\omega} \times \underline{r}$$

$$d\underline{H}_G = \underline{r} \times (\underline{\omega} \times \underline{r}) \Rightarrow \frac{d\underline{H}_G}{dm} = (x^2+y^2+z^2)\begin{bmatrix} p \\ q \\ r \end{bmatrix} - (xp+yq+zr)\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \begin{bmatrix} p(x^2+y^2+z^2) \\ q(x^2+y^2+z^2) \\ r(x^2+y^2+z^2) \end{bmatrix} - \begin{bmatrix} x^2p + xyq + xzr \\ xyp + y^2q + yzr \\ xzp + yzq + z^2r \end{bmatrix} = \begin{bmatrix} r^2+z^2 & -xy & -xz \\ -xy & x^2+z^2 & -yz \\ -xz & -yz & x^2+y^2 \end{bmatrix}\begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

$$= \begin{bmatrix} (y^2+z^2)p - xyq - xzr \\ -xyp + (x^2+z^2)q - yzr \\ -xzp - yzq + (x^2+y^2)r \end{bmatrix} = \begin{bmatrix} y^2+z^2 & -xy & -xz \\ -xy & x^2+z^2 & -yz \\ -xz & -yz & x^2+y^2 \end{bmatrix}\vec{\omega}$$

$$\therefore \underline{H}_G = \int d\underline{H}_G = \begin{bmatrix} \int y^2+z^2\,dm & -\int xy\,dm & -\int xz\,dm \\ -\int xy\,dm & \int x^2+z^2\,dm & -\int yz\,dm \\ -\int xz\,dm & -\int yz\,dm & \int x^2+y^2\,dm \end{bmatrix}$$

$\bar{x}\bar{z}$ is plane of Symmetry. $\underline{I}$ given in terms of some $\bar{x}, \bar{y}, \bar{z}$ body axes

1.)

$$I = \begin{bmatrix} I_{\bar{x}} & -I_{\bar{xy}} & -I_{\bar{xz}} \\ -I_{\bar{xy}} & I_{\bar{y}} & -I_{\bar{yz}} \\ -I_{\bar{xz}} & -I_{\bar{yz}} & I_{\bar{z}} \end{bmatrix}$$ } we are rotating the body about the $\bar{y}$ axis

$x = \bar{x}\cos\theta + \bar{z}\sin\theta$
$z = \bar{z}\cos\theta - \bar{x}\sin\theta$
$y = \bar{y}$

$$\mathcal{I} = \begin{bmatrix} \int \bar{y}^2+\bar{z}^2\,dm & -\int \bar{x}\bar{y}\,dm & -\int \bar{x}\bar{z}\,dm \\ -\int \bar{y}\bar{x}\,dm & \int \bar{x}^2+\bar{z}^2\,dm & -\int \bar{y}\bar{z}\,dm \\ -\int \bar{z}\bar{x}\,dm & -\int \bar{z}\bar{y}\,dm & \int \bar{x}^2+\bar{y}^2\,dm \end{bmatrix}$$

$$I_{xz} = \int x'z'\,dm = \int \bar{x}'\bar{z}'\,dm = \int (\bar{x}\cos\theta - \bar{z}\sin\theta)(\bar{z}\cos\theta + \bar{x}\sin\theta)\,dm$$
$$= \int \sin\theta\cos\theta(\bar{z}^2 - \bar{x}^2)\,dm + \int \cos^2\theta\,\bar{x}\bar{z}\,dm - \int \sin^2\theta\,\bar{x}\bar{z}\,dm = \cos^2\theta\, I_{\bar{xz}} + \sin\theta\cos\theta\, I_{\bar{z}} - \sin\theta\cos\theta\, I_{\bar{x}}$$

A better approach

$$\underline{\bar{v}} = \underline{I}(\underline{\bar{\omega}}) \Rightarrow \begin{bmatrix} \bar{u}_B \\ \bar{v}_B \\ \bar{w}_B \end{bmatrix} = I\begin{bmatrix} \bar{p}_B \\ \bar{q}_B \\ \bar{r}_B \end{bmatrix}$$

Additionally:

$$\begin{bmatrix} \bar{u}_B \\ \bar{v}_B \\ \bar{w}_B \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}\begin{bmatrix} u_B \\ v_B \\ w_B \end{bmatrix}$$

$$\therefore \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}\begin{bmatrix} u_B \\ v_B \\ w_B \end{bmatrix} = I\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}\begin{bmatrix} u_B \\ v_B \\ w_B \end{bmatrix}$$

$$\vec{u} = \underbrace{\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}^{-1} I \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}}_{\mathcal{I}}\vec{u}$$

$$\Rightarrow \mathcal{I} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}\begin{bmatrix} I_{\bar{x}} & -I_{\bar{xy}} & -I_{\bar{xz}} \\ -I_{\bar{xy}} & I_{\bar{y}} & -I_{\bar{yz}} \\ -I_{\bar{xz}} & -I_{\bar{yz}} & I_{\bar{z}} \end{bmatrix}\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta\, I_{\bar{x}} - \sin\theta\, I_{\bar{xz}} & -\cos\theta\, I_{\bar{xy}} - \sin\theta\, I_{\bar{yz}} & -I_{\bar{xz}}\cos\theta + I_{\bar{z}}\sin\theta \\ -I_{\bar{xy}} & I_{\bar{y}} & -I_{\bar{yz}} \\ -\sin\theta\, I_{\bar{x}} - \cos\theta\, I_{\bar{xz}} & \sin\theta\, I_{\bar{xy}} - \cos\theta\, I_{\bar{yz}} & \sin\theta\, I_{\bar{xz}} + I_{\bar{z}}\cos\theta \end{bmatrix}\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\theta\, I_{\bar{x}} + \sin^2\theta\, I_{\bar{z}} - \sin 2\theta\, I_{\bar{xz}} & -(\cos 2\theta\, I_{\bar{xy}} + \sin\theta\, I_{\bar{yz}}) & \sin^2\theta\, I_{\bar{x}} - \sin\theta\cos\theta\, I_{\bar{xz}} - I_{\bar{z}}\cos^2\theta + I_{\bar{xz}}\sin\theta\cos\theta \\ -\cos 2\theta\, I_{\bar{xy}} - \sin\theta\, I_{\bar{yz}} & I_{\bar{y}} & \sin\theta\, I_{\bar{xy}} - \cos 2\theta\, I_{\bar{yz}} \\ -\cos^2\theta\, I_{\bar{xz}} - \sin\theta\cos\theta\, I_{\bar{x}} + \sin^2\theta\, I_{\bar{xz}} + I_{\bar{z}}\sin\theta\cos\theta & I_{\bar{yz}}\sin\theta - I_{\bar{xy}}\cos\theta & \sin^2\theta\, I_{\bar{z}} + \cos^2\theta\, I_{\bar{xz}} + \sin 2\theta\, I_{\bar{xz}} \end{bmatrix}$$

2.)

Principle axes - $I_{xy} = I_{yz} = I_{xz} = 0$

Plane of Symmetry $\bar{x}\bar{z}$ $\therefore I_{\bar{xy}} = I_{\bar{yz}} = 0$

We need $I_{xz} = 0$

$$I_{xz} = (\sin^2\theta - \cos^2\theta)I_{\bar{xz}} + \tfrac{1}{2}\sin(2\theta)(I_{\bar{z}} - I_{\bar{x}})$$
$$0 = -\cos(2\theta)I_{\bar{xz}} + \tfrac{1}{2}\sin(2\theta)(I_{\bar{z}} - I_{\bar{x}})$$
$$\frac{2I_{\bar{xz}}}{I_{\bar{z}} - I_{\bar{x}}} = \tan(2\theta) \Rightarrow \theta = \tfrac{1}{2}\tan^{-1}\left(\frac{2I_{\bar{xz}}}{I_{\bar{z}} - I_{\bar{x}}}\right)$$

3.)

Δm_F, $\vec{r}_{cg} = [\bar{x}_F, \bar{y}_F, \bar{z}_F]$

Defining $\underline{I}_0 = \begin{bmatrix} I_{\bar{x}_0} & 0 & -I_{\bar{xz}_0} \\ 0 & I_{\bar{y}_0} & 0 \\ -I_{\bar{xz}_0} & 0 & I_{\bar{z}_0} \end{bmatrix}$ } with $m = m_0$ and $\omega_0 = [0\ 0\ 0]$ we need to find new cg $\bar{x}_1, \bar{z}_1$ ($y_1 = y_0$ due to Symm)

$\therefore m_0\bar{x}_0 = \bar{x}_1(m_0 - m_F) + \bar{x}_F m_F \Rightarrow \bar{x}_1 = \frac{m_0\bar{x}_0 - m_F\bar{x}_F}{m_0 - m_F}$

$\therefore \Delta\bar{x} = \bar{x}_1 - \bar{x}_0 = (\bar{x}_0 - \bar{x}_F)\frac{m_F}{m_0 - m_F}$

$\Delta\bar{z} = \bar{z}_1 - \bar{z}_0 = (\bar{z}_0 - \bar{z}_F)\frac{m_F}{m_0 - m_F}$

Parallel Axis Theorem

What is the parallel axis theorem?

Moment of inertia is just how hard it is to spin something.
Say I know the moment of inertia about an objects Cg.
Then $I_{skewed} = I_{cg} + md^2$

Knowing this:

Lets call I_{xz_0} the original I_{xz}
we can decompose it into two parts
- aircraft with no fuel } $I_{xz_{empty}}$
- fuel and fuel tanks } $I_{xz_{fuel}}$

$$I_{xz_{fuel}} = m_F((\bar{x}_0 - \bar{x}_F)^2 + \bar{y}_0^2)$$ } $I_{skewed} = I_0 + md^2$

$I_{xz_{empty}} = I_{xz_0} - I_{xz_{fuel}}$ } Gives us our I for the only aircraft about the original Cg

$I_{xz_1} = I_{xz_{empty}} - \Delta m_F(\bar{z}_1 - \bar{z}_0)^2$ } Shifts the aircraft Cg (- as we are losing mass)

Coordinate Rotation Matrices

$\underline{V}_{Earth} = [u_E, v_E, w_E]^T$

$$\underline{R}_{E1}(\psi) = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \underline{V}_{Earth} = \underline{R}_{E1}(\psi)\,\underline{V}_1$$

$$\underline{R}_{1E}(\psi) = \begin{bmatrix} \cos(\psi) & \sin(\psi) & 0 \\ -\sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \underline{V}_1 = \underline{R}_{1E}(\psi)\,\underline{V}_E$$

Step 1

$$\underline{R}_{12}(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \Rightarrow \underline{V}_1 = \underline{R}_{12}(\theta)\,\underline{V}_2$$

$$\underline{R}_{21}(\theta) = \begin{bmatrix} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \Rightarrow \underline{V}_2 = \underline{R}_{21}(\theta)\,\underline{V}_1$$

Step 2

$$\underline{R}_{2B}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix} \Rightarrow \underline{V}_2 = \underline{R}_{2B}(\phi)\,\underline{V}_B$$

$$\underline{R}_{B2}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & \sin(\phi) \\ 0 & -\sin(\phi) & \cos(\phi) \end{bmatrix} \Rightarrow \underline{V}_B = \underline{R}_{B2}(\phi)\,\underline{V}_2$$

Step 3

For some earth-fixed vector, $\underline{V}_E$, it's body-fixed representation is given by:

$$\underline{V}_B = \underline{R}_{B2}(\phi)\,\underline{R}_{21}(\theta)\,\underline{R}_{1E}(\psi)\,\underline{V}_E$$
$$= \underline{R}_{BE}(\phi, \theta, \psi)\,\underline{V}_E$$

For some body-fixed vector, $\underline{V}_B$, it's earth-fixed representation is given by:

$$\underline{V}_E = \underline{R}_{E1}(\psi)\,\underline{R}_{12}(\theta)\,\underline{R}_{2B}(\phi)\,\underline{V}_B$$
$$= \underline{R}_{EB}(\phi, \theta, \psi)\,\underline{V}_B$$

Navigation Equations

$\dot{\underline{x}}_E = [\dot{x}_E, \dot{y}_E, \dot{z}_E]^T$ (velocity in earth-fixed FOR)

$$\therefore \begin{bmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \end{bmatrix} = \underline{R}_{EB}(\phi, \theta, \psi) \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \underline{R}_{B2}(\phi)\,\underline{R}_{21}(\theta)\,R_{1E}(\psi) \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + \underline{R}_{B2}(\phi)\,\underline{R}_{21}(\theta) \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \underline{R}_{B2}(\phi) \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \end{bmatrix} = \underline{R}_{EB}(\phi, \theta, \psi) \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \underline{R}_{BE}(\phi, \theta, \psi) \begin{bmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \end{bmatrix}$$

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin(\theta) \\ 0 & \cos(\phi) & \cos(\theta)\sin(\phi) \\ 0 & -\sin(\phi) & \cos(\theta)\cos(\phi) \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin(\phi)\tan(\theta) & \cos(\phi)\tan(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi)\sec(\theta) & \cos(\phi)\sec(\theta) \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

The two navigation equations

Aircraft Dynamics on State Space

State Space Formulation

Through some manipulation we can always write out our systems as:

$$\boxed{\dot{\underline{x}}(t) = \underline{f}(\underline{x}(t), \underline{\delta})}$$

where $\underline{f}$ is an array of functions, and $\underline{x}$ is the state vector ($\underline{\delta}$ is an array of state parameters)

Examples:

$$\frac{d^2}{dt^2}[u(t)] = c\frac{d}{dt}[u(t)] + ku(t)$$

Let $v = \frac{d}{dt}(u) \Rightarrow \underline{x} = [u, v]$

Now:

$$\frac{d}{dt}\underline{x} = \frac{d}{dt}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ k & c \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix}$$

with $\underline{\delta} = (k, c)$

$$I_{xx}\dot{p}(t) - I_{xz}\dot{r}(t) = L(p, r, \delta_A, \delta_R)$$

$$I_{zz}\dot{r}(t) - I_{xz}\dot{p}(t) = N(p, r, \delta_A, \delta_R)$$

$$\frac{d}{dt}\begin{bmatrix} p \\ r \end{bmatrix} = \begin{bmatrix} I_{xx} & -I_{xz} \\ -I_{xz} & I_{zz} \end{bmatrix}^{-1}\begin{bmatrix} L(p, r, \delta_A, \delta_R) \\ N(p, r, \delta_A, \delta_R) \end{bmatrix}$$

$\delta = (\delta_A, \delta_R)$

Equilibria:

Let $\underline{x}_0$ be a singular point

Define $\underline{x} = \underline{x}_0 + \underline{x}'$, where $\underline{x}'$ is a small perturbation

$$\Rightarrow \underbrace{\frac{d}{dt}(\underline{x}_0 + \underline{x}')}_{\dot{\underline{x}}} = \underline{f}(\underline{x}_0 + \underline{x}', \underline{\delta})$$

$$\Rightarrow \frac{d}{dt}(\underline{x}_0 + \underline{x}') = \underline{f}(\underline{x}_0, \underline{\delta}) + \underline{\underline{J}}_{\underline{x}_0, \underline{\delta}}\underline{x}'$$

$$\underline{\underline{J}}_{\underline{x}_0, \underline{\delta}} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\underline{x}, \underline{\delta}) & \frac{\partial}{\partial x_2} f_1(\underline{x}, \underline{\delta}) & \cdots & \frac{\partial}{\partial x_n} f_1(\underline{x}, \underline{\delta}) \\ \vdots & & & \\ \frac{\partial}{\partial x_1} f_n(\underline{x}, \underline{\delta}) & - & - & \frac{\partial}{\partial x_n} f_n(\underline{x}, \underline{\delta}) \end{bmatrix}\Bigg|_{\underline{x} = \underline{x}_0}$$

and $\underline{f}(\underline{x}_0, \underline{\delta}) = 0 = \dot{\underline{x}}_0$ } by definition of a singular point

Thus $\dot{\underline{x}}' = \underline{\underline{J}}_{\underline{x}_0, \underline{\delta}}\underline{x}'$ } Linearised form with $\underline{x}' = \underline{x}^* e^{\lambda t}$

$$\Rightarrow \lambda \underline{x}^* e^{\lambda t} = \underline{\underline{J}}_{\underline{x}_0, \underline{\delta}}\underline{x}^* e^{\lambda t}$$

$$\Rightarrow (\underline{\underline{J}}_{\underline{x}_0, \underline{\delta}} - \underline{\underline{I}}\lambda)\underline{x}^* e^{\lambda t} = 0$$

$$|\underline{\underline{J}}_{\underline{x}_0, \underline{\delta}} - \underline{\underline{I}}\lambda| = 0$$

} We will have n solutions for λ. Will be either real or complex

Let $\underline{\xi}_1, \underline{\xi}_2, \underline{\xi}_3, \ldots, \underline{\xi}_n$ denote the set of independent eigenvectors of $\underline{\underline{J}}_{\underline{x}_0, \underline{\delta}}$

i.e. $(\underline{\underline{J}}_{\underline{x}_0, \underline{\delta}} - \underline{\underline{I}}\lambda_i)\underline{\xi}_i e^{\lambda_i t} = 0$

We can express the coordinates of any point $\underline{x}'(0)$ as $\underline{x}_0 + \underline{x}'(0) = \underline{x}_0 + k_1\underline{\xi}_1 + k_2\underline{\xi}_2 + \ldots + k_n\underline{\xi}_n$

For $t \geq 0$, $\underline{x}_0 + \underline{x}'(t) = \underline{x}_0 + k_1\underline{\xi}_1 e^{\lambda_1 t} + k_2\underline{\xi}_2 e^{\lambda_2 t} + \ldots + k_n\underline{\xi}_n e^{\lambda_n t}$

Look at what to do with repeated roots

Example

$$\frac{d}{dt}\begin{bmatrix} x \\ \dot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ k & c \end{bmatrix}\begin{bmatrix} x \\ \dot{x} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ kx + c\dot{x} \end{bmatrix}$$

$$\underline{\underline{J}} = \begin{bmatrix} \frac{\partial}{\partial x}\dot{x} & \frac{\partial}{\partial \dot{x}}\dot{x} \\ \frac{\partial}{\partial x}(kx + c\dot{x}) & \frac{\partial}{\partial \dot{x}}(kx + c\dot{x}) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ k & c \end{bmatrix}\Bigg|_{x = x_0} = \ldots$$

$f(x, \dot{x}) = 0 \Rightarrow \dot{x} = 0$, $kx + c\dot{x} = 0$ } $\underline{x}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Thus $\left|\begin{bmatrix} 0 & 1 \\ k & c \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right| = 0$

$$\begin{vmatrix} -\lambda & 1 \\ k & c - \lambda \end{vmatrix} = 0 \Rightarrow \lambda(\lambda - c) - k = 0$$

$$\lambda^2 - \lambda c - k = 0$$

$$\lambda_{1,2} = \frac{c \pm \sqrt{c^2 + 4k}}{2}$$

For simplicity, say $c = 0 \Rightarrow \lambda_{1,2} = \pm\sqrt{k}$

Then $(\underline{\underline{J}}_{\underline{x}_0, \underline{\delta}} - \underline{\underline{I}}\lambda)\underline{\xi}_i = 0$

$$\begin{bmatrix} -\sqrt{k} & 1 \\ k & -\sqrt{k} \end{bmatrix}\begin{bmatrix} v_1 \\ v_1 \end{bmatrix} = 0, \quad \begin{bmatrix} \sqrt{k} & 1 \\ k & \sqrt{k} \end{bmatrix}\begin{bmatrix} u_2 \\ v_2 \end{bmatrix} = 0$$

$v_1 - \sqrt{k}u_1 = 0$ $\quad\quad$ $\sqrt{k}u_2 + v_2 = 0$

$v_1 = \sqrt{k}u_1$ $\quad\quad$ $v_2 = -\sqrt{k}u_2$

$\therefore \underline{\xi}_1 = u_1(1, \sqrt{k})$ $\quad\quad$ $\underline{\xi}_2 = u_2(1, -\sqrt{k})$

Now finally state space trajectory through $\underline{x}' = [0, 1]^T$ | i.e. $\dot{x}'$ now equals 1

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = k_1\begin{bmatrix} 1 \\ \sqrt{k} \end{bmatrix} + k_2\begin{bmatrix} 1 \\ -\sqrt{k} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ \sqrt{k} & -\sqrt{k} \end{bmatrix}\begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \frac{1}{-2\sqrt{k}}\begin{bmatrix} -\sqrt{k} & -1 \\ -\sqrt{k} & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{-2\sqrt{k}}\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2\sqrt{k}} \\ -\frac{1}{2\sqrt{k}} \end{bmatrix} = \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}$$

Thus:

$$\begin{bmatrix} x \\ \dot{x} \end{bmatrix} = \frac{1}{2\sqrt{k}}\begin{bmatrix} 1 \\ \sqrt{k} \end{bmatrix}e^{\sqrt{k}t} - \frac{1}{2\sqrt{k}}\begin{bmatrix} 1 \\ -\sqrt{k} \end{bmatrix}e^{-\sqrt{k}t}$$

$$\Rightarrow x = \frac{1}{2\sqrt{k}}\left(e^{\sqrt{k}t} - e^{-\sqrt{k}t}\right)$$

$$\dot{x} = \frac{1}{2}\left(e^{\sqrt{k}t} - e^{-\sqrt{k}t}\right)$$

$\dot{\underline{x}}' = \begin{bmatrix} 0 & 1 \\ k & c \end{bmatrix}\begin{bmatrix} x' \\ \dot{x}' \end{bmatrix} = \underline{\underline{J}}_{x_0, \dot{x}_0}\underline{x}'$

already $\underline{\underline{J}}_{x_0, \dot{x}_0}$ as no dependence on x or $\dot{x}$

$$\left|\underline{\underline{J}} - \underline{\underline{I}}\lambda\right| = 0 \Rightarrow \begin{vmatrix} -\lambda & 1 \\ k & c - \lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - \lambda c - k = 0$$

$$\Rightarrow \lambda_{1,2} = \frac{c \pm \sqrt{c^2 + 4k}}{2}$$

Case 1: $c^2 + 4k > 0 \Rightarrow \lambda_{1,2} = \frac{c \pm \sqrt{c^2 + 4k}}{2}$

$\underline{\xi}_1 = \begin{bmatrix} -\lambda_1 & 1 \\ k & c - \lambda_1 \end{bmatrix} \Rightarrow \underline{\xi}_1 = (1, \frac{c + \sqrt{c^2 + 4k}}{2})u_1$

$\underline{\xi}_2 = \begin{bmatrix} -\lambda_2 & 1 \\ k & c - \lambda_2 \end{bmatrix} \Rightarrow \underline{\xi}_2 = (1, \frac{c - \sqrt{c^2 + 4k}}{2})u_2$

Case 2: $c^2 + 4k = 0 \Rightarrow \lambda_{1,2} = \frac{c}{2}$

$\therefore \underline{\xi}_1 = u_1(1, \frac{c}{2})$

$\begin{bmatrix} -\frac{c}{2} & 1 \\ k & -\frac{c}{2} \end{bmatrix}\underline{\xi}_2 = \underline{\xi}_1 = \begin{bmatrix} 1 \\ \frac{c}{2} \end{bmatrix}$

$-\frac{c}{2}\xi_{21} + \xi_{22} = 1 \Rightarrow \xi_{22} = 1 + \frac{c}{2}\xi_{21}$

$k\xi_{21} + \frac{c}{2}\xi_{22} = \frac{c}{2}$

$k\xi_{21} + \frac{c}{2}(1 + \frac{c}{2}\xi_{21}) = \frac{c}{2}$

$\xi_{21}(k + \frac{c^2}{4}) = 0$

Case 3: $c^2 + 4k < 0 \Rightarrow$ Let $w = -(c^2 + 4k)$

$\Rightarrow \lambda_{1,2} = \frac{c}{2} \pm \frac{i\sqrt{w}}{2}$

$\Rightarrow \begin{bmatrix} -\frac{c}{2} - i\frac{\sqrt{w}}{2} & 1 \\ k & -\frac{c}{2} + i\frac{\sqrt{w}}{2} \end{bmatrix}\begin{bmatrix} \xi_{11} \\ \xi_{12} \end{bmatrix} = 0 \Rightarrow \underline{\xi}_1 = (1, \frac{c}{2} + i\frac{\sqrt{w}}{2})$

$\Rightarrow \underline{\xi}_2 = (0, 1)$

$\Rightarrow \begin{bmatrix} -\frac{c}{2} + i\frac{\sqrt{w}}{2} & 1 \\ k & \frac{c}{2} + i\frac{\sqrt{w}}{2} \end{bmatrix}\begin{bmatrix} \xi_{21} \\ \xi_{22} \end{bmatrix} = 0 \Rightarrow \underline{\xi}_2 = (1, \frac{c}{2} - i\frac{\sqrt{w}}{2})$

$\underline{x}' = k_1\underline{\xi}_1 e^{\lambda_1 t} + k_2\underline{\xi}_2 e^{\lambda_2 t}$ | For some perturbation, we solve for k_1, k_2 at $t = 0$

$\Rightarrow \underline{x}'_{t=0} = k_1\underline{\xi}_1 + k_2\underline{\xi}_2 = \begin{bmatrix} \underline{\xi}_1 & \underline{\xi}_2 \end{bmatrix}\begin{bmatrix} k_1 \\ k_2 \end{bmatrix} \Rightarrow \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} \underline{\xi}_1 & \underline{\xi}_2 \end{bmatrix}^{-1}\underline{x}'_{t=0}$

Structure of Singular Points

Dividing up the n eigenvalues $(\lambda_1, \lambda_2, \ldots, \lambda_n)$ into the
- p real eigenvalues $(\lambda_1, \lambda_2, \ldots, \lambda_p) \Rightarrow$ eigenvectors $\underline{\xi}_1, \underline{\xi}_2, \ldots, \underline{\xi}_p$
- q pairs of complex $(\lambda_1^+, \lambda_1^-, \ldots, \lambda_q^+, \lambda_q^-) \Rightarrow$ eigenvectors $\underline{\xi}_1^+, \underline{\xi}_1^-, \ldots, \underline{\xi}_q^+, \underline{\xi}_q^-$

$$\Rightarrow \underline{x}_0 + \underline{x}'(t) = \underline{x}_0 + k_1\underline{\xi}_1 e^{\lambda_1 t} + k_2\underline{\xi}_2 e^{\lambda_2 t} + \ldots + k_p\underline{\xi}_p e^{\lambda_p t} + k_1^+\underline{\xi}_1^+ e^{\lambda_1^+ t} + k_2^+\underline{\xi}_2^+ e^{\lambda_2^+ t} + \ldots + k_q^+\underline{\xi}_q^+ e^{\lambda_q^+ t} + k_1^-\underline{\xi}_1^- e^{\lambda_1^- t} + k_2^-\underline{\xi}_2^- e^{\lambda_2^- t} + \ldots + k_q^-\underline{\xi}_q^- e^{\lambda_q^- t}$$ } Form of $\underline{x}'(t)$

$$= x_0 + k_1\underline{\xi}_1 e^{\lambda_1 t} + \ldots + k_p\underline{\xi}_p e^{\lambda_p t} + k_1^+\underline{\xi}_1^+ e^{(r_1 + is_1)t} + \ldots + k_q^+\underline{\xi}_q^+ e^{(r_q + is_q)t} + k_1^-\underline{\xi}_1^- e^{(r_1 - is_1)t} + \ldots + k_q^-\underline{\xi}_q^- e^{(r_q - is_q)t}$$ } Letting $\lambda_j^+ = r_j + is_j$, $\lambda_j^- = r_j - is_j$

$$= x_0 + \ldots + k_p\underline{\xi}_p e^{\lambda_p t} + k_1^+\underline{\xi}_1^+ e^{r_1 t}e^{is_1 t} + k_1^-\underline{\xi}_1^- e^{r_1 t}e^{-is_1 t} + \ldots + k_q^+\underline{\xi}_q^+ e^{r_q t}e^{is_q t} + k_q^-\underline{\xi}_q^- e^{r_q t}e^{-is_q t}$$ } Grouping the complex components

$$= x_0 + \ldots + k_p\underline{\xi}_p e^{\lambda_p t} + k_1^+\underline{\xi}_1^+ e^{r_1 t}(\cos(s_1 t) + i\sin(s_1 t)) + k_1^-\underline{\xi}_1^- e^{r_1 t}(\cos(s_1 t) - i\sin(s_1 t)) + \ldots + k_q^+\underline{\xi}_q^+ e^{r_q t}(\cos(s_q t) + i\sin(s_q t)) + k_q^-\underline{\xi}_q^- e^{r_q t}(\cos(s_q t) - i\sin(s_q t))$$ } Using euler's formula

$$= x_0 + \ldots + k_p\underline{\xi}_p e^{\lambda_p t} + k_1^+\underline{\xi}_1^+ e^{r_1 t}\left(\underline{\xi}_1^+ + \left(\frac{k_1^-}{k_1^+}\right)\underline{\xi}_1^-\right)\cos(s_1 t) + k_1^+ e^{r_1 t}\left(\underline{\xi}_1^+ - \frac{k_1^-}{k_1^+}\underline{\xi}_1^-\right)i\sin s_1 t + \ldots$$

$$= x_0 + \ldots + k_p\underline{\xi}_p e^{\lambda_p t} + k_q' e^{r_q t}\left(\underline{\xi}_q^+\cos(s_q t) + \underline{\xi}_q^+\sin(s_q t)\right)$$ } factorising with $k_j'\underline{\xi}_j^+ = k_j^+\left[\underline{\xi}_j^+ + \left(\frac{k_j^-}{k_j^+}\right)\underline{\xi}_j^-\right]$, $k_j'\underline{\xi}_j^- = k_j^+\left[\underline{\xi}_j^+ - \frac{k_j^-}{k_j^+}\underline{\xi}_j^-\right]$

Basically we can turn off every other k_j's and just look at the way an isolated mode impacts the dynamics.

The Aircraft's State Equations

What we have so far: (everything is in body fixed)

Note: Φ, Θ, Ψ are euler-angles, α, Θ, γ are body-fixed angles

$$\Rightarrow \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = R_{E_B}(\Phi, \Theta, \Psi) \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} p + q\sin\Phi\tan\Theta + r\cos\Phi\tan\Theta \\ q\cos\Phi - r\sin\Phi \\ q\sin\Phi\sec\Theta + r\cos\Phi\sec\Theta \end{bmatrix}$$

$$\left.\begin{aligned} m(\dot{u} + Vr - Wq) &= X^a + mg_x \\ m(\dot{v} + Ur - Vp) &= Y^a + mg_y \\ m(\dot{w} - Uq + Vp) &= Z^a + mg_z \end{aligned}\right\}$$ We split the net forces X,Y,Z into the aerodynamic ($X^a, \dots$) and gravitational ($mg_x, \dots$) components

$$\begin{bmatrix} g_x \\ g_y \\ g_z \end{bmatrix} = R_{BE}(\phi, \theta, \psi) \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} = \begin{bmatrix} -g\sin(\theta) \\ g\cos(\theta)\sin(\phi) \\ g\cos(\theta)\cos(\phi) \end{bmatrix}$$

$$\left.\begin{aligned} I_{xx}\dot{p} - I_{xz}\dot{r} + (I_{zz} - I_{yy})qr - I_{xz}pq &= \mathcal{L}^a \\ I_{yy}\dot{q} + (I_{xx} - I_{zz})pr + I_{xz}(p^2 - r^2) &= M^a \\ I_{zz}\dot{r} - I_{xz}\dot{p} + (I_{yy} - I_{xx})pq + I_{xz}qr &= N^a \end{aligned}\right\}$$ our moment ODE's

$$\left.\begin{aligned} p + q\sin\Phi\tan\Theta + r\cos\Phi\tan\Theta &= \dot{\Phi} \\ q\cos\Phi - r\sin\Phi &= \dot{\Theta} \\ q\sin\Phi\sec\Theta + r\cos\Phi\sec\Theta &= \dot{\Psi} \end{aligned}\right\}$$ relating our euler angles to our body fixed rotation rates

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State Space Formulation

State Vector: $\underline{x} = (U, W, q, \Theta, V, p, r, \Phi, \Psi)$

Standard Form: $\underline{\mathcal{M}}\,\dot{\underline{x}} = \underline{N}(\underline{x}) + \underline{F}(\dot{\underline{x}}, \underline{x}) + \underline{G}(\underline{x})$

$$\underline{\mathcal{M}}\,\dot{\underline{x}} = \begin{bmatrix} m & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & m & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I_{yy} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & m & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I_{xx} & -I_{xz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -I_{xz} & I_{zz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{U} \\ \dot{W} \\ \dot{q} \\ \dot{\Theta} \\ \dot{V} \\ \dot{p} \\ \dot{r} \\ \dot{\Phi} \\ \dot{\Psi} \end{bmatrix}$$ (Inertial Matrix) $\cdot\, \dot{x}$

$$\underline{N}(\underline{x}) = \begin{bmatrix} m(Vr - Wq) \\ m(Uq - Vp) \\ (I_{zz} - I_{xx})pr - I_{xz}(p^2 - r^2) \\ q\cos\Phi - r\sin\Phi \\ m(Wp - Ur) \\ (I_{yy} - I_{zz})qr + I_{xz}pq \\ (I_{xx} - I_{yy})pq - I_{xz}qr \\ p + q\sin\Phi\tan\Theta + r\cos\Phi\tan\Theta \\ q\sin\Phi\sec\Theta + r\cos\Phi\sec\Theta \end{bmatrix}$$ Nonlinear Matrix

$$\underline{F}(\dot{\underline{x}}, \underline{x}) = \begin{bmatrix} X^a(\dot{\underline{x}}, \underline{x}) \\ Z^a(\dot{\underline{x}}, \underline{x}) \\ M^a(\dot{\underline{x}}, \underline{x}) \\ 0 \\ Y^a(\dot{\underline{x}}, \underline{x}) \\ \mathcal{L}^a(\dot{\underline{x}}, \underline{x}) \\ N^a(\dot{\underline{x}}, \underline{x}) \\ 0 \\ 0 \end{bmatrix}$$ Forces Matrix (Aerodynamic + Propulsive forces)

$$\underline{G}(\underline{x}) = \begin{bmatrix} -mg\sin\Theta \\ mg\cos\Theta\cos\Phi \\ 0 \\ 0 \\ mg\cos\Theta\sin\Phi \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$ Gravitational Matrix

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Analysis

$\dot{\underline{x}} = 0 \Rightarrow \underline{N}(\underline{x}_0) + \underline{F}(0, \underline{x}_0) + \underline{G}(\underline{x}_0) = 0$ ($\underline{x}_0$ is the equilibrium state)

definition of a singular point

Letting $\underline{x} = \underline{x}_0 + \underline{x}'$ ($\frac{d}{dt}\underline{x} = \frac{d}{dt}\underline{x}_0 + \frac{d}{dt}\underline{x}' = \dot{\underline{x}}'$, with $\frac{d}{dt}\underline{x}_0 = 0$)

$$\underline{\mathcal{M}}(\dot{\underline{x}}_0^{\,0} + \dot{\underline{x}}') = \underline{N}(\underline{x}_0 + \underline{x}') + \underline{F}(\dot{\underline{x}}', \underline{x}_0 + \dot{\underline{x}}') + \underline{G}(\underline{x}_0 + \underline{x}')$$

By the Taylor expansion ($A_{\underline{x}_0}$ = Jacobian of $A(\underline{x})$ evaluated at $\underline{x}_0$)

$$\underline{\mathcal{M}}\,\dot{\underline{x}}' = \underline{N}(\underline{x}_0) + \underline{N}_{\underline{x}_0}\underline{x}' + \underline{F}(0, \underline{x}_0) + \underline{F}_{\underline{x}_0}\underline{x}' + \underline{F}_{\dot{\underline{x}}_0}\dot{\underline{x}}' + \underline{G}(\underline{x}_0) + \underline{G}_{\underline{x}_0}\underline{x}'$$

$\underline{N}(\underline{x}_0) + \underline{F}(0, \underline{x}_0) + \underline{G}(\underline{x}_0) = 0$ by equilibrium

$$\Rightarrow \underline{\mathcal{M}}\,\dot{\underline{x}}' = \underline{N}_{\underline{x}_0}\underline{x}' + \underline{F}_{\underline{x}_0}\underline{x}' + \underline{F}_{\dot{\underline{x}}_0}\dot{\underline{x}}' + \underline{G}_{\underline{x}_0}\underline{x}'$$

Evaluating at $\underline{x}_0, \dot{\underline{x}} = 0$ (so all $\dot{\underline{x}}$ terms vanish)

Defining $\underline{\mathcal{M}}_* = \underline{\mathcal{M}} - \underline{F}_{\dot{\underline{x}}_0}$, the "augmented inertia matrix"

$$\Rightarrow \underline{\mathcal{M}}_*\,\dot{\underline{x}}' = (\underline{N}_{\underline{x}_0} + \underline{F}_{\underline{x}_0} + \underline{G}_{\underline{x}_0})\,\underline{x}'$$

Example

$$\frac{d^2x}{dt^2} + c\frac{d}{dt}x + k\sin x = 0$$

$$\Rightarrow \frac{d}{dt}\begin{bmatrix} \dot{x} \\ x \end{bmatrix} = \begin{bmatrix} -c\dot{x} - k\sin x \\ \dot{x} \end{bmatrix}$$

$$\Rightarrow \underline{f}(\dot{\underline{x}}, \underline{x}) = \begin{bmatrix} -c\dot{x} - k\sin x \\ \dot{x} \end{bmatrix} \Rightarrow -c\dot{x} - k\sin x = 0 \quad \dot{x} = 0$$

$$\therefore \dot{x}_0 = 0, \; x_0 = n\pi$$

$$\underline{J} = \begin{bmatrix} -c & -k\cos x \\ 1 & 0 \end{bmatrix}\Big|_{\underline{x} = \underline{x}_0} = \begin{bmatrix} -c & (-1)^{n+1}k \\ 1 & 0 \end{bmatrix}$$

Thus $\dot{\underline{x}}' = \begin{bmatrix} -c & (-1)^{n+1}k \\ 1 & 0 \end{bmatrix}\underline{x}'$

$$|\underline{J} - \lambda\underline{I}| = \begin{vmatrix} -c - \lambda & (-1)^{n+1}k \\ 1 & -\lambda \end{vmatrix}$$

$$= \lambda(\lambda + c) - (-1)^{n+1}k$$

$$= \lambda^2 + c\lambda + (-1)^n k$$

$$\Rightarrow \lambda_{1,2} = \frac{-c \pm \sqrt{c^2 - 4k(-1)^n}}{2}$$

$$= -\frac{c}{2} \pm \sqrt{\left(\frac{c}{2}\right)^2 - k(-1)^n}$$

Wings Level Trimmed Flight

During this course we will only consider trimmed steady flight, wings level, with no sideslip.

Trim Conditions:

Consider $x_0 = (U_0, W_0, q_0, \theta_0, V_0, p_0, r_0, \Phi_0, \Psi_0)$

having the particular constrained form:

Trimmed Flight Spec:

- $U_0 = U_e$, $V_0 = 0$, $W_0 = 0$ } Ensures no-sideslip, x-axis aligned to direction of travel. The orientation of the B-F axes depends on U_e
- $p_0 = 0$, $q_0 = 0$, $r_0 = 0$ } No angular acceleration
- $\Phi_0 = 0$ } Ensures no bank angle to E-F horizontal
- $\theta_0 = \theta_e$ } direction of motion is inclined by θ_e to E-F horizontal
- $\Psi_0 = 0$ } Implies our E-F axes are aligned in $\hat{i}$ to the B-F axes

Trim Equations:

Subbing x_0 into our $\underline{N}$, $\underline{F}$, and $\underline{G}$ matrices:

$$\underline{F}(\underline{x}_0, \underline{u}_0) = \begin{bmatrix} X_e^a \\ Z_e^a \\ M_e^a \\ 0 \\ Y_e^a \\ L_e^a \\ N_e^a \\ 0 \\ 0 \end{bmatrix}, \quad \underline{G}(\underline{x}_0) = \begin{bmatrix} -mg\sin\theta_e \\ mg\cos\theta_e \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \underline{N}(\underline{x}_0) = \underline{0}$$

Thus:

$$\begin{bmatrix} X_e^a - mg\sin\theta_e \\ Z_e^a + mg\cos\theta_e \\ M_e^a \\ 0 \\ Y_e^a \\ L_e^a \\ N_e^a \\ 0 \\ 0 \end{bmatrix} = 0$$

Linearised Form

$$\underline{x} = (U, W, q, \theta, V, p, r, \Phi, \Psi)$$

After taking the Jacobian and subbing $\underline{x}_0$:

$$\underline{N}_{\underline{x}_0} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & mU_e & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -mU_e & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & \tan\theta_e & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \sec\theta_e & 0 & 0 \end{bmatrix}, \quad \underline{G}_{\underline{x}_0} = \begin{bmatrix} 0 & 0 & 0 & -mg\cos\theta_e & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -mg\sin\theta_e & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & mg\cos\theta_e & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$\underline{F}_{\underline{x}_0} = \begin{bmatrix} X_u & X_w & X_q & X_\theta & X_v & X_p & X_r & X_\Phi & X_\Psi \\ Z_u & Z_w & Z_q & Z_\theta & Z_v & Z_p & Z_r & Z_\Phi & Z_\Psi \\ M_u & M_w & M_q & M_\theta & M_v & M_p & M_r & M_\Phi & M_\Psi \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ Y_u & Y_w & Y_q & Y_\theta & Y_v & Y_p & Y_r & Y_\Phi & Y_\Psi \\ L_u & L_w & L_q & L_\theta & L_v & L_p & L_r & L_\Phi & L_\Psi \\ N_u & N_w & N_q & N_\theta & N_v & N_p & N_r & N_\Phi & N_\Psi \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$ replace each subscript with a dot and/or to get $\underline{F}_{\dot{\underline{x}}_0}$

$\underline{F}_{\underline{x}_0}$ and $\underline{F}_{\dot{\underline{x}}_0}$ are the stability derivatives of the aircraft, $(X_u, Y_{\dot{u}}, M_{u,\dots})$ $\left[X_u = \left.\frac{\partial \dot{x}^a}{\partial u}\right|_{\underline{x}=\underline{x}_0, \dot{\underline{x}}=0} \right]$

Simplifying the linearised form:

1.) We can set all $\dot{\Phi}, \dot{\theta}, \dot{\Psi}$ derivatives to 0 as they only represent a change in coordinates, not conditions

2.) We can neglect all unsteady ($\dot{u}$) derivatives but $X_{\dot{u}}, M_{\dot{w}}$, and $Z_{\dot{w}}$

3.) We can split the symmetric (U, W, q, θ) and asymmetric states (V, p, r, Φ, Ψ), which we can split forces into (X^a, Z^a, M^a) and (Y^a, L^a, N^a)

4.) We assume the pitch and yaw force are indep. of orientation and position (only true at altitude)

Thus assuming this all holds (esp. the Curie Principle):

The Longitudinal, Symmetric equations of motion:

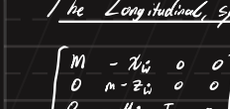
$$\begin{bmatrix} m - X_{\dot{u}} & 0 & 0 \\ 0 & m - Z_{\dot{w}} & 0 \\ 0 & -M_{\dot{w}} & I_{yy} \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} X_u & X_w & X_q & -mg\cos\theta_e \\ Z_u & Z_w & Z_q + mU_e & -mg\sin\theta_e \\ M_u & M_w & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix}$$

The Lateral, Asymmetric equations of motion:

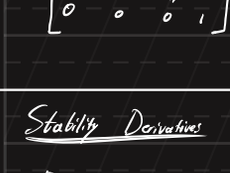
$$\begin{bmatrix} m & 0 & 0 & 0 & 0 \\ 0 & I_{xx} & -I_{xz} & 0 & 0 \\ 0 & -I_{xz} & I_{zz} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{p} \\ \dot{r} \\ \dot{\Phi} \\ \dot{\Psi} \end{bmatrix} = \begin{bmatrix} Y_v & Y_p & Y_r - mU_e & mg\cos\theta_e & 0 \\ L_v & L_p & L_r & 0 & 0 \\ N_v & N_p & N_r & 0 & 0 \\ 0 & 1 & \tan\theta_e & 0 & 0 \\ 0 & 0 & \sec\theta_e & 0 & 0 \end{bmatrix} \begin{bmatrix} v \\ p \\ r \\ \Phi \\ \Psi \end{bmatrix}$$

Stability Derivatives

Equilibrium:



Perturbed:



Longitudinal Derivatives

T_e

$$X^a = \overbrace{T\cos(\epsilon+\alpha_e)} - \tfrac{1}{2}\rho V_{e0}^2 S_{ref}\left[C_{D_0}\cos(\alpha-\alpha_e) - C_L\sin(\alpha-\alpha_e)\right]$$

$-T_e$

$$Z^a = \overbrace{-T\sin(\epsilon+\alpha_e)} - \tfrac{1}{2}\rho V_{e0}^2 S_{ref}\left[C_L\cos(\alpha-\alpha_e) + C_{D_0}\sin(\alpha-\alpha_e)\right]$$

$$M^a = T\ell_T + \tfrac{1}{2}\rho V_{e0}^2 S_{ref}\bar{c}\left[C_{M_{e0}} - K_m C_L\right] \Big\} \begin{matrix} \ell_T = (x_{cg} - x_T)\sin(\epsilon+\alpha_e) \\ \quad + (z_{cg} - z_T)\cos(\epsilon+\alpha_e) \end{matrix}$$

(+ve inward axis, and 0 the ref. moment of 0)

$$V_{e0} = U_e + \frac{\partial V_{e0}}{\partial u}u' + \frac{\partial V_{e0}}{\partial v}v' + \frac{\partial V_{e0}}{\partial w}w' \quad \text{| taylor}$$

$$\Leftarrow V_{e0}^2 = (U_e+u')^2 + v'^2 + w'^2 \quad \text{| Pythagoras}$$

$$\Rightarrow U_e^2 + 2U_e\frac{\partial V_{e0}}{\partial u}u' + 2U_e\frac{\partial V_{e0}}{\partial v}v' + 2U_e\frac{\partial V_{e0}}{\partial w}w' = U_e^2 + 2U_e u' \quad \text{| combining}$$

$$\Rightarrow \boxed{\frac{\partial V_{e0}}{\partial u} = 1, \quad \frac{\partial V_{e0}}{\partial v} = 0, \quad \frac{\partial V_{e0}}{\partial w} = 0}$$

$$\alpha = \alpha_e + \frac{\partial\alpha}{\partial u}u' + \frac{\partial\alpha}{\partial w}w' \quad \text{| taylor}$$

$$\alpha = \alpha_e + \tan^{-1}\left(\frac{w'}{U_e+u'}\right) \approx \alpha_e + \frac{w'}{U_e} \quad \text{| trig + approximation}$$

$$\Rightarrow \boxed{\frac{\partial\alpha}{\partial u} = 0, \quad \frac{\partial\alpha}{\partial v} = 0, \quad \frac{\partial\alpha}{\partial w} = \frac{1}{U_e}}$$

$$C_L = C_{L_{0w}} + a_w\alpha + \frac{S_H}{S_{ref}}\left[C_{L_{0H}} + a_H(1-\tfrac{d\epsilon}{d\alpha})\alpha + a_E\delta_E + a_T\delta_T\right]$$

$$C_D = C_{D_0} + \frac{1}{\pi AR_w e_w}(C_{L_0} + a_w\alpha)^2 + \frac{S_H}{S_{ref}}\left[\frac{1}{\pi AR_H e_H}\left[C_{L_{0H}} + a_H(1-\tfrac{d\epsilon}{d\alpha})\alpha\right]^2 + d_E\delta_E + d_T\delta_T\right]$$

$$M^a = T\ell_T + \tfrac{1}{2}\rho V_{e0}^2 S_{ref}\bar{c}\left\{C_{M_{0w}} + \frac{x_{cg}-x_w}{\bar{c}}(C_{L_{0w}} + a_w\alpha) + \frac{S_H(x_{cg}-x_H)}{S_{ref}\bar{c}}\left[C_{L_{0H}} + a_H\alpha(1-\tfrac{d\epsilon}{d\alpha})\alpha + a_E\delta_E + a_T\delta_T\right]\right\}$$

This comes from saying $C_M = C_{M_0} + \frac{\partial C_{M_0}}{\partial C_L}C_L$, $K_n = \frac{x_{cg}-x_{ac}}{\bar{c}}$, $\frac{\partial C_{M_0}}{\partial C_L} = -K_n$ ← this can be shown

Control

Look at Gebistor, what's important is that I know how changing properties (eg Sref, C_L, dihedral, etc) affects the Control Stability derivatives.

U - Derivatives (X_u, Z_u, M_u)

$$\frac{\partial x^u}{\partial u} = \frac{\partial x^u}{\partial \alpha}\frac{\partial \alpha}{\partial u} + \frac{\partial x^u}{\partial V_{e0}}\frac{\partial V_{e0}}{\partial u} \approx \frac{\partial x^u}{\partial V_{e0}}$$
$$\frac{\partial z^u}{\partial u} = \frac{\partial z^u}{\partial \alpha}\frac{\partial \alpha}{\partial u} + \frac{\partial z^u}{\partial V_{e0}}\frac{\partial V_{e0}}{\partial u} \approx \frac{\partial z^u}{\partial V_{e0}}$$
$$\frac{\partial M^u}{\partial u} = \frac{\partial M^u}{\partial \alpha}\frac{\partial \alpha}{\partial u} + \frac{\partial M^u}{\partial V_{e0}}\frac{\partial V_{e0}}{\partial u} \approx \frac{\partial M^u}{\partial V_{e0}}$$

if there was also powerplant induced forces we would need an extra $\frac{\partial T}{\partial T_c}\cdot\frac{\partial T}{\partial u}$ term

$\therefore$

$$\frac{\partial x^u}{\partial u} = \frac{\partial \bar{T}_x}{\partial V_{e0}} - \rho V_{e0} S_{ref}\left[C_D\cos(\alpha-\alpha_0) - C_L\sin(\alpha-\alpha_0)\right] - \tfrac{1}{2}\rho V_{e0}^2 S_{ref}\left[\frac{\partial C_D}{\partial V_{e0}}\cos(\alpha-\alpha_0) - \frac{\partial C_L}{\partial V_{e0}}\sin(\alpha-\alpha_0)\right]$$
$$= \left(\frac{\partial T_x}{\partial V_{e0}}\right)_c - \rho U_e S_{ref}(C_D)_c - \tfrac{1}{2}\rho U_e^2\left(\frac{\partial C_D}{\partial V_{e0}}\right)_c \quad \Big| \begin{array}{l}(\cdot)_c \text{ means calculated} \\ \text{at } \alpha=\alpha_0,\ V_{e0}=U_e\end{array}$$

easy to do for the others

The effect of velocity on aero coeffs. is typically a result of compress. effects so it's better to write the derivatives in terms of Mach number

$$\Rightarrow \frac{\partial}{\partial V_{e0}} = \frac{dM_{e0}}{dV_{e0}}\frac{\partial}{\partial M_{e0}} = \frac{1}{c_s}\frac{\partial}{\partial M_{e0}}$$

$\therefore$

$$X_u = \frac{1}{c_s}\left(\frac{\partial T_x}{\partial M_{e0}}\right)_c - \rho U_e S_{ref}(C_D)_c - \overbrace{\frac{\rho U_e^2 S_{ref}}{2c_s}}^{\text{term always present}}\overbrace{\left(\frac{\partial C_D}{\partial M_{e0}}\right)_c}^{\text{only really relevant in transonic/supersonic speeds}}$$

$$Z_u = -\frac{1}{c_s}\left(\frac{\partial T_z}{\partial M_{e0}}\right)_c - \rho U_e S_{ref}(C_L)_c - \frac{\rho U_e^2 S_{ref}}{2c_s}\left(\frac{\partial C_L}{\partial M_{e0}}\right)_c$$

$$M_u = \underbrace{\frac{L_T}{c_s}\left(\frac{\partial T}{\partial M_{e0}}\right)_c + \rho U_e S_{ref}\bar{c}\left[C_{m_0} - K_n(C_L)_c\right]}_{\text{if } L_T=0,\ \text{then } C_{m_0} - K_n(C_L)_c = 0 \text{ as well}} + \frac{\rho U_e^2 S_{ref}\bar{c}}{2c_s}\Bigg[\underbrace{\left(\frac{\partial C_m}{\partial M_{e0}}\right)_c}_{\text{represents compressibility and elasticity effects}} - \underbrace{\frac{\partial k_n}{\partial M_{e0}}}_{\text{at supersonic mach this is big but at transonic it won't be as the aero centre is shifting}}(C_L)_c - K_n\left(\frac{\partial C_L}{\partial M_{e0}}\right)_c\Bigg]$$

NOT UNCOMMON FOR M_u TO BE SMALL OR EVEN 0

W - Derivatives (X_w, Z_w, M_w)

$$\frac{\partial x^u}{\partial w} = \frac{\partial x^u}{\partial \alpha}\frac{\partial \alpha}{\partial w} + \frac{\partial x^u}{\partial V_{e0}}\frac{\partial V_{e0}}{\partial w} \approx \frac{1}{U_e}\frac{\partial x^u}{\partial \alpha}$$
$$\frac{\partial z^u}{\partial w} = \frac{\partial z^u}{\partial \alpha}\frac{\partial \alpha}{\partial w} + \frac{\partial z^u}{\partial V_{e0}}\frac{\partial V_{e0}}{\partial w} \approx \frac{1}{U_e}\frac{\partial z^u}{\partial \alpha}$$
$$\frac{\partial M^u}{\partial w} = \frac{\partial M^u}{\partial \alpha}\frac{\partial \alpha}{\partial w} + \frac{\partial M^u}{\partial V_{e0}}\frac{\partial V_{e0}}{\partial w} \approx \frac{1}{U_e}\frac{\partial M^u}{\partial \alpha}$$

W doesn't really represent speeds just angle of attack changes (we want to keep sensible derivatives)

$\therefore$

$$X_w = \frac{1}{U_e}\left(\frac{\partial T_x}{\partial \alpha}\right)_e + \tfrac{1}{2}\rho U_e S_{ref}\left[(C_L)_c - \left(\frac{\partial C_D}{\partial \alpha}\right)_c\right]$$
$$Z_w = -\frac{1}{U_e}\left(\frac{\partial T_z}{\partial \alpha}\right)_e - \tfrac{1}{2}\rho U_e S_{ref}\left[\left(\frac{\partial C_L}{\partial \alpha}\right)_c + (C_D)_c\right]$$
$$M_w = \frac{L_T}{U_e}\left(\frac{\partial T}{\partial \alpha}\right)_e - \tfrac{1}{2}\rho U_e S_{ref}\bar{c}\,k_n\left(\frac{\partial C_L}{\partial \alpha}\right)_c$$

$\frac{\partial C_D}{\partial \alpha} = \frac{2}{\pi AR e}(C_L)_c\frac{\partial C_L}{\partial \alpha} = \frac{2a}{\pi AR e}(C_L)_c$ (if we assume a parabolic drag polar)

From $M_w = \frac{1}{2}\rho U_e S_{ref}\bar{c}\left(\frac{\partial C_m}{\partial \alpha}\right)_c$ assuming C_{m_0} is constant and assuming $\frac{dL_T}{d\alpha}=0$

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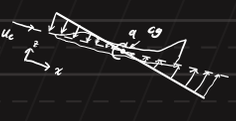
q - Derivatives (X_q, Z_q, M_q)

in this case the eqn but in general we always use the body axes not geo.



$U_i = -(z-z_0)q$, $W_i = (x-x_0)q$

We do each component separately w.r.t x_i, z being the B-F axes coordinates of the aerodynamic centre

$$\frac{\partial x}{\partial q} = \frac{\partial x}{\partial U}\frac{\partial U_i}{\partial q} + \frac{\partial x}{\partial W}\frac{\partial W_i}{\partial q} = -(z-z_0)\frac{\partial x}{\partial U} + (x-x_0)\frac{\partial x}{\partial W}$$
$$\frac{\partial z}{\partial q} = \frac{\partial z}{\partial U}\frac{\partial U_i}{\partial q} + \frac{\partial z}{\partial W}\frac{\partial W_i}{\partial q} = -(z-z_0)\frac{\partial z}{\partial U} + (x-x_0)\frac{\partial z}{\partial W}$$
$$\frac{\partial M}{\partial q} = \frac{\partial M}{\partial U}\frac{\partial U_i}{\partial q} + \frac{\partial M}{\partial W}\frac{\partial W_i}{\partial q} + (x-x_0)\frac{\partial z}{\partial q} - (z-z_0)\frac{\partial x}{\partial q} = -(z-z_0)\left(\frac{\partial M}{\partial U} + \frac{\partial x}{\partial q}\right) + (x-x_0)\left(\frac{\partial M}{\partial W} + \frac{\partial z}{\partial q}\right)$$

A major contribution is the tailplane and since the position from the cg $U_i \approx 0$

$$(X_q)_H = \left(\frac{\partial x_H^a}{\partial w}\right)_c = \tfrac{1}{2}\rho\eta_H U_e S_H\left[(C_{L_H})_c - \left(\frac{\partial C_{D_H}}{\partial \alpha}\right)_c\right]$$
$$(Z_q)_H = \left(\frac{\partial z_H^a}{\partial w}\right)_c = -\tfrac{1}{2}\rho\eta_H U_e S_H\left[\left(\frac{\partial C_{L_H}}{\partial \alpha}\right)_c + (C_{D_H})_c\right]$$
$$(M_q)_H = \left(\frac{\partial M_H}{\partial w}\right)_c = \tfrac{1}{2}\rho\eta_H U_e S_H\bar{c}_H\left(\frac{\partial C_{m_H}}{\partial \alpha}\right)_c \Big\} \text{ most cases } (M_q)_H = 0$$

$\therefore$

$$(X_q)_H \approx \tfrac{1}{2}\rho\eta_H U_e S_H l_H'\left[(C_{L_H})_0 - \left(\frac{\partial C_{D_H}}{\partial \alpha}\right)_c\right]$$
$$(Z_q)_H \approx -\tfrac{1}{2}\rho\eta_H U_e S_H l_H'\left[\left(\frac{\partial C_{L_H}}{\partial \alpha}\right)_c + (C_{D_H})_c\right]$$
$$(M_q)_H \approx l_H'(z_H)_H - h_H(x_H)_H + \left(\frac{\partial x_H^a}{\partial q}\right)_c \approx l_H'^2(z_H)_H - h_H l_H'(x_H)_H + l_H'(M_H)_H$$

where:
$l_H' = x_H - x_0$,
$h_H = z_H - z_0$,
taken positive when the aero centre is aft and above cg respectively

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$\dot{W}$ - Derivatives

Derivation is kind of involved so just note we assume the tailplane is the big contributor:

$$X_{\dot{w}} = \frac{L_H}{U_e}\left(\frac{dL}{d\alpha}\right)_e(x_w)_H$$
$$Z_{\dot{w}} = \frac{L_H}{U_e}\left(\frac{dL}{d\alpha}\right)_e(z_w)_H$$
$$M_{\dot{w}} = \frac{L_H}{U_e}\left(\frac{dL}{d\alpha}\right)_e\left[-h_H(x_w)_H + l_H'(z_w)_H + (M_w)_H\right]$$